

RudinAnalysis

†NEURON†

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# Chapter 1

## Overview

This blueprint records the current formal development of `RudinAnalysis`. The code follows the chapter order of Rudin's *Principles of Mathematical Analysis*, starting with ordered fields and the construction of the real numbers, then moving to metric topology and the theory of numerical sequences.

The project is intentionally elementary in style: many results are reproved directly from the chosen axioms so that the formal development mirrors the book as closely as possible.

**Definition 1.** The present formalization is organized into three mathematical blocks: ordered algebraic structures, basic topology of metric spaces, and convergence theory for sequences and subsequences.

## Chapter 2

# The Real and Complex Number Systems

### 2.1 Ordered sets and completeness

**Definition 2.** For  $E \subseteq \alpha$  and  $u, \ell \in \alpha$ :

$$u \text{ is an upper bound of } E \iff \forall x \in E, x \leq u,$$

$$\ell \text{ is a lower bound of } E \iff \forall x \in E, \ell \leq x.$$

Accordingly,

$$E \text{ is bounded above} \iff \exists u, \forall x \in E, x \leq u,$$

$$E \text{ is bounded below} \iff \exists \ell, \forall x \in E, \ell \leq x.$$

**Definition 3.** For  $E \subseteq \alpha$  and  $s, i \in \alpha$ :

$$s = \sup E \iff (\forall x \in E, x \leq s) \wedge (\forall u, (\forall x \in E, x \leq u) \rightarrow s \leq u),$$

$$i = \inf E \iff (\forall x \in E, i \leq x) \wedge (\forall \ell, (\forall x \in E, \ell \leq x) \rightarrow \ell \leq i).$$

**Definition 4.** The completeness predicates are:

$$\forall E \subseteq \alpha, E \neq \emptyset \wedge E \text{ bounded above} \rightarrow \exists s, s = \sup E,$$

$$\forall E \subseteq \alpha, E \neq \emptyset \wedge E \text{ bounded below} \rightarrow \exists i, i = \inf E.$$

**Theorem 5.** For a linear order,

$$LUB \text{ property} \iff GLB \text{ property}.$$

**Theorem 6.** Any type equipped with the project's completeness structure satisfies the greatest-lower-bound property.

**Theorem 7.** Every nonempty set that is bounded above admits a supremum.

**Theorem 8.** Every nonempty set that is bounded below admits an infimum.

## 2.2 Field axioms and ordered fields

**Definition 9.** The project introduces a custom structure of field axioms, close to Rudin's presentation, in order to rederive elementary algebraic laws from first principles.

**Theorem 10.** *Within any type satisfying these field axioms, one has additive and multiplicative cancellation laws.*

**Definition 11.** An ordered field is a linear order together with the project's field axioms and the expected compatibility of order with addition and multiplication by positive elements.

**Theorem 12.** *In an ordered field, negatives of positive elements are negative, squares of nonzero elements are positive, and inversion reverses order on positive elements.*

## 2.3 Irrationality and Dedekind reals

**Theorem 13.** *The formal development proves that there is no rational number whose square is equal to 2.*

**Definition 14.** The real numbers are constructed as Dedekind cuts of rational numbers.

**Theorem 15.** *The ordered set of Dedekind reals satisfies the completeness axiom used elsewhere in the project.*

**Theorem 16.** *Addition on Dedekind reals is commutative and associative, and every element has an additive inverse.*

**Theorem 17.** *The Dedekind reals carry the expected ordered-field operations: multiplication is commutative and associative, distributes over addition, and every nonzero element admits a multiplicative inverse.*

## Chapter 3

# Basic Topology

### 3.1 Metric preliminaries

**Definition 18.** For a metric space  $(X, d)$ ,  $x \in X$ , and  $r > 0$ :

$$B(x, r) = \{y \in X : d(x, y) < r\},$$

$$\bar{B}(x, r) = \{y \in X : d(x, y) \leq r\},$$

$$B^*(x, r) = \{y \in X : 0 < d(x, y) < r\}.$$

**Definition 19.** The basic notions are encoded as:

$$E \text{ bounded} \iff \exists x \in X \exists R > 0, E \subseteq B(x, R),$$

$$p \text{ is a limit point of } E \iff \forall r > 0, (B^*(p, r) \cap E) \neq \emptyset,$$

$$(x_n) \text{ bounded} \iff \exists x \in X \exists R > 0, \forall n, x_n \in B(x, R).$$

**Theorem 20.** For  $E \subseteq X$ ,

$$E \text{ is open} \iff \forall x \in E, \exists r > 0, B(x, r) \subseteq E.$$

**Theorem 21.** Every open ball is an open set.

**Theorem 22.** Every deleted ball is open.

**Theorem 23.** If  $p$  is a limit point of  $E$ , then

$$\forall r > 0, (B(p, r) \cap E) \text{ is infinite.}$$

### 3.2 Compactness

**Definition 24.** The project packages the finite-subcover language needed to compare Rudin's compactness with the compactness API from `mathlib`.

**Theorem 25.** For  $K \subseteq X$ ,

$$K \text{ compact} \iff \forall \{U_i\}_{i \in I}, K \subseteq \bigcup_{i \in I} U_i \wedge (\forall i, U_i \text{ open}) \rightarrow \exists i_1, \dots, i_n, K \subseteq U_{i_1} \cup \dots \cup U_{i_n}.$$

**Theorem 26.** *Compactness behaves well under induced topologies, and compact subsets of metric spaces are closed.*

**Theorem 27.** *For a compact set, any family of closed subsets with the finite intersection property has nonempty total intersection.*

**Theorem 28.** *Every infinite subset of a compact metric space has a limit point in that compact set.*

## Chapter 4

# Numerical Sequences and Series

### 4.1 Convergent sequences

**Definition 29.** For a sequence  $(x_n)$  in a metric space and  $p \in X$ ,

$$x_n \rightarrow p \iff \forall \varepsilon > 0, \exists N, \forall n \geq N, d(x_n, p) < \varepsilon.$$

The blueprint also uses the bundled predicates “ $(x_n)$  is convergent” and “ $\lim x_n$ ”.

**Theorem 30.** For a sequence  $(x_n)$  and  $p \in X$ ,

$$x_n \rightarrow p \iff \forall r > 0, \exists N, \forall n \geq N, x_n \in B(p, r).$$

**Theorem 31.** Limits of sequences in metric spaces are unique.

**Theorem 32.**

$$x_n \rightarrow p \implies (x_n) \text{ is bounded.}$$

**Theorem 33.** If  $p$  is a limit point of  $E$ , then there exists a sequence  $(x_n)$  such that

$$\forall n, x_n \in E, \quad x_n \neq p, \quad x_n \rightarrow p.$$

**Theorem 34.** For complex-valued sequences,

$$x_n \rightarrow x, y_n \rightarrow y \implies x_n + y_n \rightarrow x + y, \quad x_n y_n \rightarrow xy,$$

$$x_n \rightarrow x, c \in \mathbb{C} \implies cx_n \rightarrow cx, \quad x_n + c \rightarrow x + c.$$

### 4.2 Subsequences and compactness

**Definition 35.** A subsequence of  $(x_n)$  is given by a strictly increasing map

$$\phi : \mathbb{N} \rightarrow \mathbb{N}, \quad \phi(m) < \phi(n) \text{ whenever } m < n,$$

and the induced sequence is  $(x_{\phi(n)})$ .

**Theorem 36.** If  $x_n \rightarrow p$  and  $\phi$  is strictly increasing, then

$$x_{\phi(n)} \rightarrow p.$$

**Theorem 37.** *If  $(x_n)$  is a sequence in a compact metric space, then there exist a strictly increasing  $\phi : \mathbb{N} \rightarrow \mathbb{N}$  and  $p$  such that*

$$x_{\phi(n)} \rightarrow p.$$

*In particular, every bounded sequence in  $\mathbb{R}^m$  has a convergent subsequence.*

**Definition 38.** The set of subsequential limits of  $(x_n)$  is

$$\text{SubLim}(x_n) = \left\{ p \in X : \exists \phi : \mathbb{N} \rightarrow \mathbb{N} \text{ strictly increasing, } x_{\phi(n)} \rightarrow p \right\}.$$

**Theorem 39.** *The set of subsequential limits of a sequence is closed.*

### 4.3 Cauchy sequences and completeness

**Definition 40.** For a sequence  $(x_n)$  in a metric space:

$$(x_n) \text{ is Cauchy} \iff \forall \varepsilon > 0, \exists N, \forall m, n \geq N, d(x_m, x_n) < \varepsilon.$$

Its  $N$ -th tail set is

$$T_N = \{x_n : n \geq N\}.$$

Completeness is expressed by

$$X \text{ complete} \iff \text{every Cauchy sequence in } X \text{ converges.}$$

**Theorem 41.**

$$(x_n) \text{ is Cauchy} \iff \text{diam}(T_N) \rightarrow 0.$$

Moreover,

$$\text{diam}(\overline{E}) = \text{diam}(E).$$

**Theorem 42.** *If  $(K_n)$  is a decreasing sequence of nonempty compact sets with*

$$K_{n+1} \subseteq K_n \quad \text{and} \quad \text{diam}(K_n) \rightarrow 0,$$

*then*

$$\bigcap_{n=0}^{\infty} K_n = \{p\}$$

*for some point  $p$ .*

**Theorem 43.**

$$x_n \rightarrow p \implies (x_n) \text{ is Cauchy.}$$

Also,

$$X \text{ compact} \implies \text{every Cauchy sequence in } X \text{ converges,}$$

*and in particular every Cauchy sequence in  $\mathbb{R}^m$  converges.*

**Theorem 44.**

$$X \text{ compact} \implies X \text{ complete,}$$

$$\mathbb{R}^m \text{ is complete,}$$

$$X \text{ complete and } E \subseteq X \text{ closed} \implies E \text{ complete.}$$



## 4.4 Monotone sequences

**Definition 45.** For a real sequence  $(s_n)$ :

increasing  $\iff \forall m \leq n, s_m \leq s_n$ ,

decreasing  $\iff \forall m \leq n, s_n \leq s_m$ ,

monotone  $\iff$  increasing or decreasing.

**Theorem 46.** For a monotone real sequence  $(s_n)$ ,

$(s_n)$  converges  $\iff (s_n)$  is bounded.